

Resolved-record many-body Lyapunov analysis at four circumferences: hard and soft adaptive chiral domain walls

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We analyze 600 independent maximally mixed Gaussian Born trajectories of the adaptive chiral circuit at $N_x = 20$, $N_y = 20, 30, 40$, and depth $T = 4N_y$. The hard support-truncated arm and corrected soft untruncated arm each contain $S = 100$ trajectories per circumference. The saved one-particle occupations fix the normalized many-body singular spectrum, while the cumulative realized Born log probability restores its absolute normalization. All 120 result and completion pairs pass the server-pinned byte-count, SHA-256, configuration, source, event-count, and numerical checks. The leading rate λ_0 is stable between $[2N_y, 3N_y]$ and $[3N_y, 4N_y]$ at every size, and its same-window agreement with the record-weight slope is better than 1.8×10^{-5} relatively. This repairs the temporal failure of the earlier $2N_y$ pilot. It does not, however, produce a controlled CFT coefficient: the three-size $1/N_y^2$ coefficients have bootstrap intervals containing zero and order-unity sensitivity to fit form. Moreover, late soft-wall excited levels are mostly below the double-precision occupation resolution. The data establish absolute-spectrum reconstruction, leading-rate convergence, and wall-localized residual purification, but not a precision c_{eff} or labeled operator spectrum.

I. QUESTION AND COMPLETED ENSEMBLE

For a resolved measurement record $\mathbf{m}$, the monitored circuit defines a many-body evolution operator (mEO) $\hat{V}_{\mathbf{m}}(t)$. We ask whether the new $T = 4N_y$ purification ensemble is deep enough to extract its leading many-body Lyapunov rates and the finite-size coefficients used by Zabalo *et al.* [arXiv:2107.03393v3 (2022)]. This is a Lyapunov spectrum of an operator product, not a Lieb–Robinson velocity or light-cone spectrum.

The locked protocol is $N_x = 20$, $N_y = 20, 30, 40$, maximally mixed initialization, $n_{\text{shell}} = 1$, $\alpha_1 = 1$, $\alpha_2 = 30$, raster- y measurement order, perfect correction, no postselection, and complex128 dynamics through the canonical GPU entry point `classA_U1FGTN_gpu.run_markov_circuit`. Each construction contains $S = 100$ independent Born trajectories at every N_y . Table I keeps the two immutable execution identities explicit.

V3 changes only continuation-time representation selection. A heterogeneous covariance batch may contain trajectories already at projector precision and others that remain mixed; v3 keeps the explicitly requested covariance representation across that restart. The measurement protocol, Born rule, seeds, observer, and complex128 engine are unchanged. Hard-v2 completed before that continuation failure; soft-v3 is the corrected completed soft ensemble.

TABLE I. Completed inputs. A result pair is one five-trajectory NPZ and its checksum-binding completion JSON. The two revisions are compared scientifically but are not rewritten or silently pooled.

arm	revision	pairs	trajectories
hard, support truncated	v2	60	300
soft, untruncated	v3	60	300

II. ABSOLUTE GAUSSIAN SPECTRUM

For the normalized conditional output from the maximally mixed probe,

$$\hat{\rho}_{R,\mathbf{m}}^V(t) = \frac{\hat{V}_{\mathbf{m}}(t)\hat{V}_{\mathbf{m}}^\dagger(t)}{Z_{\mathbf{m}}(t)}, \quad (1)$$

the one-particle occupations $\nu_a^{\mathbf{m}}(t)$ generate the normalized many-body weights

$$w_{\mathbf{n}}^{\mathbf{m}}(t) = \prod_a [\nu_a^{\mathbf{m}}(t)]^{n_a} [1 - \nu_a^{\mathbf{m}}(t)]^{1-n_a}. \quad (2)$$

The recorded cumulative Born log probability

$$\omega_{\mathbf{m}}(t) = \log p_{\mathbf{m}}(t) \quad (3)$$

restores the absolute normalization,

$$\log Z_{\mathbf{m}}(t) = N_{\text{eff}} \log 2 + \omega_{\mathbf{m}}(t). \quad (4)$$

Here $N_{\text{eff}} = 22N_y$ for the hard arm, whose record starts after the one-time Born-conditioned exterior preparation, and $N_{\text{eff}} = 40N_y$ for the soft arm, whose record starts from the global maximally mixed state.

We use the squared-singular-value convention

$$\ell_i^{\mathbf{m}}(t) = \log[\sigma_i^{\mathbf{m}}(t)]^2 = t\lambda_i^{\mathbf{m}} + o(t). \quad (5)$$

The leading value and finite excitation costs are

$$\ell_0^{\mathbf{m}}(t) = \log Z_{\mathbf{m}}(t) + \sum_a \log \max(\nu_a, 1 - \nu_a), \quad (6)$$

$$d_a(t) = \left| \log \frac{\nu_a(t)}{1 - \nu_a(t)} \right|. \quad (7)$$

The leading 64 levels are generated from the smallest subset sums of d_a ; no exponentially large Fock matrix is formed. Modes at an exact occupation cap have infinite flip cost. We use 10^{-9} only as the locked numerical

resolution inherited from the production occupation guard. A level below that resolution is censored, not replaced by a clipped finite gap.

At $t = 0$, both transfer problems give $\ell_i = 0$. For the hard arm the full saved spectrum additionally contains deterministic exterior product caps; these contribute neither to Eq. (4) nor to the finite leading levels.

III. VERIFICATION AND INFERENCE PROTOCOL

The analysis rehashes all 240 imported files against the immutable Drive inventory. It then verifies revision-specific result and completion schemas, configuration hashes, all four executed-source hashes, task seeds, sample indices, cycle grids, event counts, float64 record accumulation, occupation bounds, contour closure, and the final complex128 covariance. Exactly 600 sample indices pass. The maximum record accumulation error is 1.82×10^{-12} ; the largest entropy and charge closure errors are 2.27×10^{-13} and 1.42×10^{-14} . The largest occupation-bound roundoff is 4.78×10^{-10} , below its 10^{-9} guard, and the final covariance Hermiticity residual is zero at stored precision.

Every trajectory is fit separately in the fixed windows

$$W_1 = [N_y, 2N_y], \quad W_2 = [2N_y, 3N_y], \quad W_3 = [3N_y, 4N_y]. \quad (8)$$

Only after fitting do we average over trajectories. Temporal acceptance compares paired W_2 and W_3 estimates from the same record. A quantity passes when its 2000-replicate whole-trajectory bootstrap interval contains zero and the relative shift is below 10%. At least 95% of trajectories must contain a numerically resolved quantity throughout both windows. Hard and soft records are independent and are never paired with one another.

IV. RESULTS

A. Leading rate and record closure

The leading rate passes the $W_2 \rightarrow W_3$ temporal gate for all six construction-size groups. The relative shifts at $N_y = 20, 30, 40$ are 1.32%, 0.62%, and 0.25% for hard walls and 0.73%, 0.52%, and 0.46% for soft walls. Thus $4N_y$, unlike the earlier $2N_y$ endpoint, supplies a stable late window for λ_0 .

The independent, checksum-pinned hard-wall $T = 2N_y$ analysis gave leading-rate decisions fail, fail, and pass at $N_y = 20, 30, 40$; the present $T = 4N_y$ hard arm gives pass, pass, and pass. The 800- and 300-trajectory ensembles are compared only as a depth diagnostic and are never pooled.

The absolute normalization closes much more tightly. In W_3 the relative difference between $d\omega/dt$ and λ_0 is

1.76×10^{-5} , 7.60×10^{-6} , and 5.65×10^{-6} for the hard arm, and 1.86×10^{-6} , 1.87×10^{-7} , and 1.14×10^{-7} for the soft arm. The record slope is therefore the leading singular-value slope to far better than the preregistered 1% closure criterion. Born-generated records receive an ordinary arithmetic ensemble mean; applying a second Born weight would be incorrect.

B. Censored excited spectrum

The longer evolution creates a precision boundary for the excited spectrum. In the hard arm the first three gaps remain resolved for all trajectories in W_2 and W_3 , whereas the fourth is resolved for only 83–86%. In the soft arm the first gap is resolved for 44–49%, the second and third for only 2–6%, and the fourth for none. This is consistent with the much more complete soft-wall purification, but it prevents an unbiased late-window soft gap fit in complex128. Fitting only the surviving less-purified records would condition on the outcome and bias the ensemble, so the missing values are reported as censored.

For the hard arm, Δ_3 is the only ordered gap passing the complete temporal, resolution, positive-coefficient, and 20% finite-size sensitivity gates. Its descriptive coefficient is

$$A_3 = 4.827 [4.453, 5.194], \quad \alpha x_3 = 0.768 [0.709, 0.827]. \quad (9)$$

This is a provisional ordered-gap coefficient, not an identified operator dimension. The files contain occupation eigenvalues but not their natural eigenvectors, momentum, charge-sector, or two-wall symmetry labels. The index “3” therefore denotes sorted order only.

C. Finite-size coefficient

Following Eqs. (3) and (4) of Zabalo *et al.*, we fit

$$-\frac{\lambda_0}{N_y} = f_\infty + \frac{A_0}{N_y^2}, \quad (10)$$

$$\frac{\lambda_0 - \lambda_i}{N_y} = \frac{A_i}{N_y^2}. \quad (11)$$

The leading hard-wall result is

$$A_0 = +5.59 [-15.19, 26.98], \quad (12)$$

and the soft-wall result is

$$A_0 = -6.13 [-29.21, 17.20]. \quad (13)$$

Both intervals contain zero. Adding the exactly determined $1/N_y^4$ term changes A_0 to 59.7 and -51.7 , while omitting $N_y = 20$ changes it to 30.7 and -27.2 . These are sensitivity factors of 9.70 and 7.45 relative to the primary coefficients, far above 20%. The formal diagnostics $\alpha c_{\text{eff}} = -10.7$ and $+11.7$ are consequently rejected rather

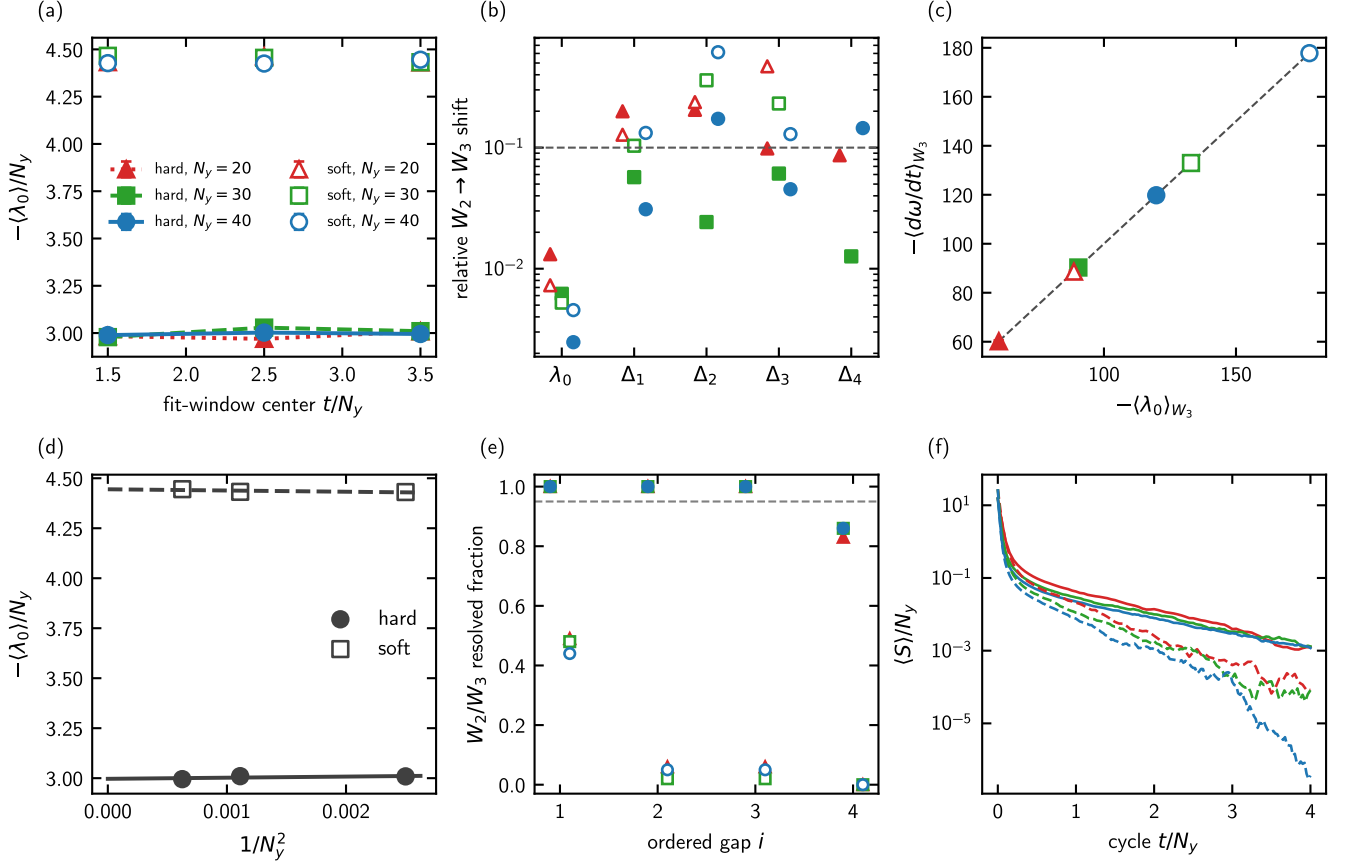


FIG. 1. Many-body Lyapunov and purification analysis. Each point uses $S = 100$ independent maximally mixed Born trajectories at fixed N_y ; lines and filled symbols denote hard walls and open symbols or dashed lines denote soft walls. Error bars in (a) and (c) are trajectory standard errors. (a) Window-resolved leading free-energy density. (b) Magnitude of the paired $W_2 \rightarrow W_3$ shift; the dashed line is the 10% magnitude condition, while acceptance additionally requires a paired 95% bootstrap interval containing zero and 95% numerical resolution. (c) Same- W_3 record-weight and leading spectral slopes; the dashed line is equality. (d) Three-size $1/N_y^2$ fits. (e) Fraction of trajectories numerically resolving each ordered gap through both W_2 and W_3 ; the dashed line is the 95% requirement. (f) Total entropy per circumference through $T = 4N_y$. Fits are trajectory-first and bootstrap intervals use 2000 whole-trajectory replicates.

than reported as physical estimates. Three circumferences cannot separate a universal $1/N_y^2$ term from subleading structure at the observed noise level.

Random without-replacement $S = 25, 50, 75$ subsets retain broad, sign-changing A_0 distributions. Increasing samples alone is therefore not the obvious next repair; additional circumferences or an independently constrained finite-size form are needed.

D. Purification and spatial residuals

At $4N_y$, the mean hard-wall entropy is 0.0249, 0.0393, and 0.0474 at $N_y = 20, 30, 40$; the corresponding charge variances are 0.00706, 0.0113, and 0.0134. Approximately 96.2–96.5% of both residuals lies in the fixed three-column windows around $x = 5$ and $x = 15$. The soft-wall endpoint entropy is 1.67×10^{-3} , 2.46×10^{-3} , and 1.47×10^{-5} ; its

charge variance is 3.77×10^{-4} , 5.87×10^{-4} , and 1.25×10^{-6} . About 94.8–95.6% of the soft charge variance remains in the wall windows. The soft entropy fraction is less stable because its denominator is already nearly zero. These contours localize residual mixedness; they do not localize individual singular modes.

V. CLAIM LEDGER AND INTERPRETATION

The completed analysis supports four statements. First, the Gaussian occupation spectrum and record weight reconstruct an absolute many-body singular spectrum with controlled normalization. Second, λ_0 reaches a stable late-time window by $4N_y$. Third, residual entropy and charge variance are concentrated near the two interfaces. Fourth, one hard-wall ordered-gap coefficient survives the preregistered numerical gates as a descriptive quantity.

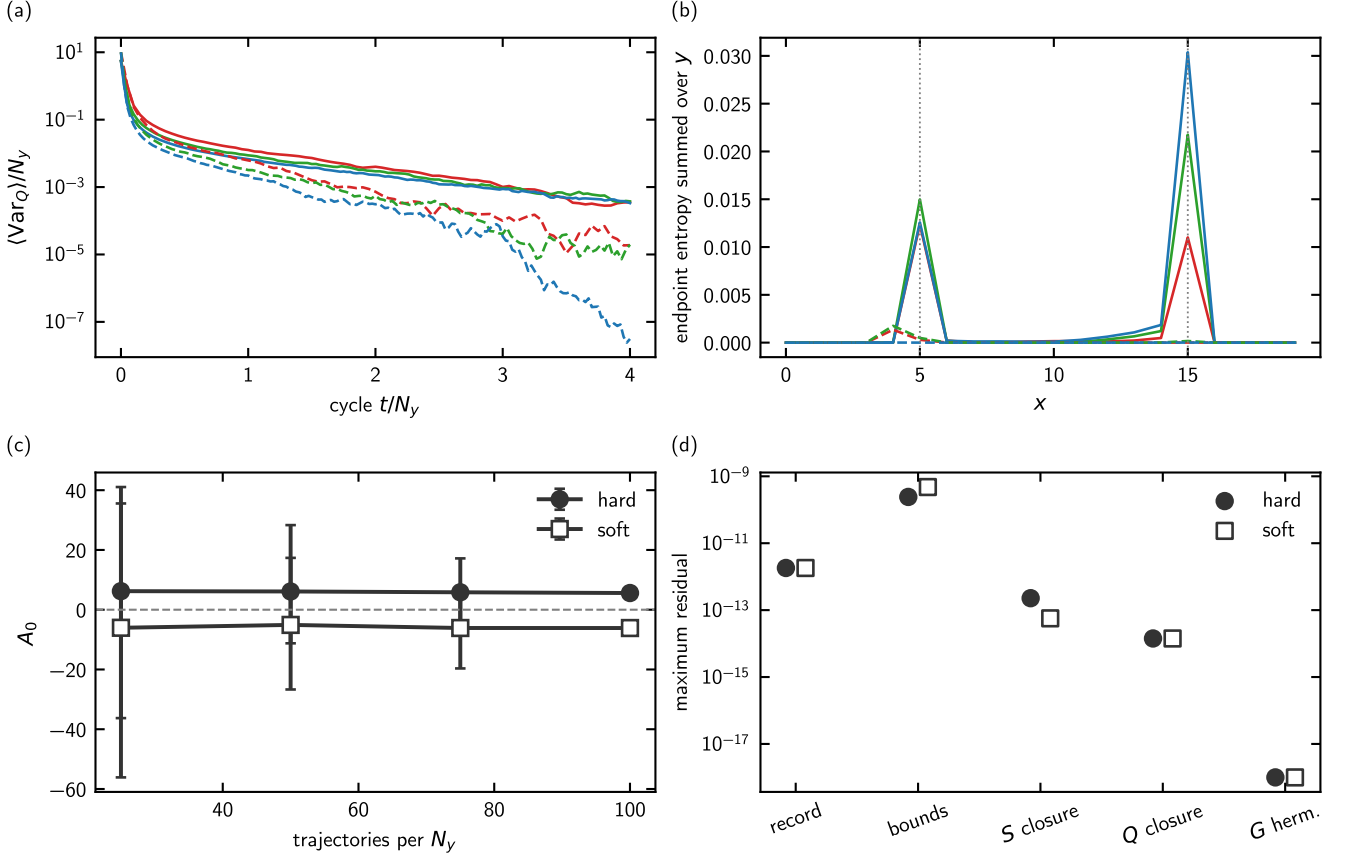


FIG. 2. Numerical and purification diagnostics for the same $S = 100$, $T = 4N_y$ independent-trajectory ensembles. (a) Total intrinsic charge variance per circumference. (b) Endpoint entropy profiles summed along the periodic direction; dotted vertical lines mark the domain walls at $x = 5, 15$. (c) Random without-replacement sample convergence of A_0 ; error bars for $S < 100$ are central 95% intervals across 200 subsets, while $S = 100$ is the unique full ensemble. (d) Maximum record, occupation-bound, entropy-closure, charge-closure, and final-covariance Hermiticity residuals. Colors encode $N_y = 20, 30, 40$ as red triangles, green squares, and blue circles.

It does not support a precision c_{eff} , absolute x_i , or a conformal tower assignment. The anisotropy α is not independently calibrated, the circumference grid has only three points, and the soft excited spectrum is precision-censored. The already established spatial evidence for boundary criticality is useful context, but it cannot rescue an unstable independent Lyapunov finite-size coefficient. The discrepancy is a finite-size and numerical-resolution limitation, not a failure of the boundary-critical state.

VI. REPRODUCIBILITY

The machine-readable directory accompanying this note contains trajectory window slopes, temporal and record-closure ledgers, finite-size and sample-size fits, cycle summaries, endpoint profiles, shardwise numerical diagnostics, and `analysis_summary.json`. The analysis command is

```
python analyze_completed_campaign.py
```

from bundle 07. Production acceptance requires the default checksum pass; the development-only `--skip-file-hashes` option is not used for the final report. The bootstrap seed is 2026090907.

VII. CONCLUSION

The $4N_y$ data answer the primary depth question. The leading many-body Lyapunov rate and absolute record normalization have converged at all three circumferences for both hard and soft walls. The remaining obstruction is not trajectory count or leading-time convergence: it is the narrow three-size grid for A_0 and, for soft-wall excitations, double-precision censoring after near complete purification. The defensible paper result is therefore a validated absolute Gaussian spectrum and stable leading rate, accompanied by a clear negative result for precision CFT coefficient extraction.